Baseline Raman Suppression and Core Optimization Surface

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note is the front door to the project. It explains the baseline single-mode, phase-only Raman-suppression experiment: choose an input spectral phase, propagate the pulse through a nonlinear fiber model, and minimize the fraction of output energy that lands in the Raman-shifted band. The canonical SMF-28 run shown here improves the Raman-band objective from $-1.11\,\mathrm{dB}$ without shaping to $-44.79\,\mathrm{dB}$ after optimization, while passing the numerical trust checks. The baseline is not the deepest path in the repo. Its purpose is to define the physics, objective, diagnostics, and vocabulary that all later research directions modify.

Claim Status

Status: established reference workflow
The single-mode phase-only workflow is stable enough to serve as the common baseline for the research-note series. The strongest claim is methodological: the project has a coherent objective, adjoint-gradient optimizer, no-optimization control, standard image bundle, and trust-report gate. Deeper optimization strategies are covered in separate notes.

1 Question and Thesis

The baseline question is whether phase-only input shaping can suppress Raman-band energy in a clean, reproducible single-mode fiber experiment.

The answer is yes. The maintained baseline changes only the spectral phase $\phi(\omega)$, keeps the input amplitude fixed, propagates through a single-mode nonlinear fiber model, and minimizes the output Raman-band energy fraction. That gives a deliberately narrow reference problem:

$$\text{phase-only control} \quad \Longrightarrow \quad \text{lower Raman-band output fraction.}$$

This matters because every later path changes one piece of this setup. Reduced-basis work changes the allowed phase shapes. Sharpness work changes the objective so local robustness matters. Trust-region work changes the optimizer. Long-reach and multimode work change the physical model. This note defines the common object those paths start from.

Intuition Check

Think of the pulse as many colors traveling together. Raman scattering moves energy toward longer wavelengths. The phase mask tries to rearrange the timing of those colors before the fiber so the nonlinear evolution sends less energy into the unwanted Raman-shifted region.

2 Physical Problem

The canonical benchmark in this note is

$$\text{SMF-28}, \qquad L = 2\,\text{m}, \qquad P = 0.2\,\text{W}, \qquad \lambda_0 = 1550\,\text{nm}.$$

The maintained run uses $N_t = 8192$, a 27 ps time window, and a 185 fs input pulse. The fiber propagation follows the generalized nonlinear Schrödinger equation, with dispersion, Kerr nonlinearity, and Raman response as in standard nonlinear-fiber optics [1, 2]. The numerical propagation style follows interaction-picture / split-step ideas used for supercontinuum and nonlinear pulse simulations [3].

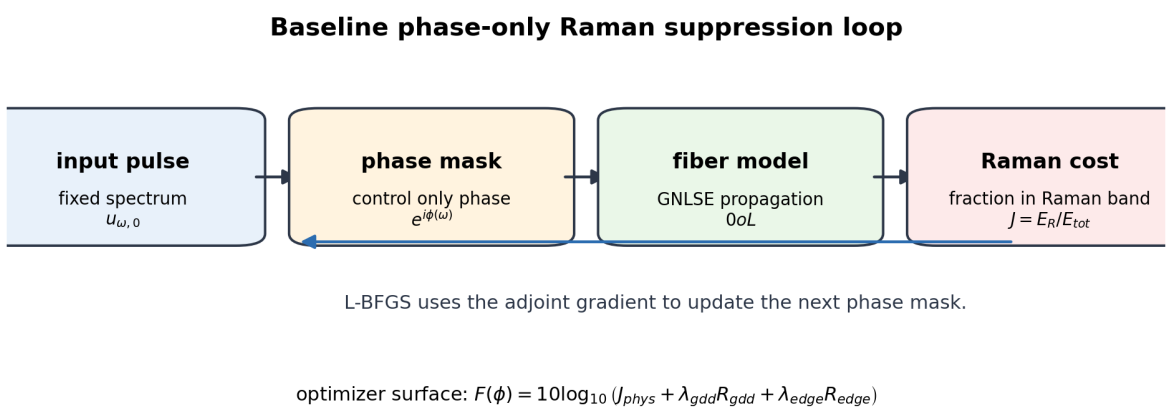


Figure 1: Baseline phase-only optimization loop. The optimizer proposes a spectral phase, the nonlinear fiber solver propagates the shaped pulse, the Raman-band output fraction is measured, and the adjoint gradient tells L-BFGS how to update the phase.

The shaped input spectrum is

$$u_\omega(0;\phi) = u_{\omega,0}\exp(i\phi(\omega)).$$

Only phase is changed. The spectral amplitude is not intentionally reduced, so the optimizer cannot win by throwing away input energy. This is why the result is physically meaningful as a phase-shaping result rather than an attenuation trick.

Intuition Check

Amplitude shaping can suppress unwanted output partly by deleting input power. Phase shaping is a stricter test: the same colors are still present, but their relative timing is changed.

3 The Objective

At the fiber output, define the Raman-band energy and total output energy:

$$E_{\mathcal{B}_{\mathrm{Raman}}}(\phi) = \sum_{\omega_k \in \mathcal{B}_{\mathrm{Raman}}} |u_{\omega_k}(L;\phi)|^2 \, \Delta\omega, \qquad E_{\mathrm{tot}}(\phi) = \sum_k |u_{\omega_k}(L;\phi)|^2 \, \Delta\omega.$$

The physics cost is the fraction

$$J_{\mathrm{phys}}(\phi) = \frac{E_{\mathcal{B}_{\mathrm{Raman}}}(\phi)}{E_{\mathrm{tot}}(\phi)}.$$

Lower is better. The reported value is usually

$$J_{\mathrm{dB}}(\phi) = 10 \log_{10} J_{\mathrm{phys}}(\phi).$$

For example, $-40\,\mathrm{dB}$ means roughly 10^{-4} of the output energy lies in the Raman band.

The maintained optimizer uses a regularized linear surface before applying the dB transform:

$$J_{\mathrm{surf}}(\phi) = J_{\mathrm{phys}}(\phi) + \lambda_{\mathrm{gdd}} R_{\mathrm{gdd}}(\phi) + \lambda_{\mathrm{edge}} R_{\mathrm{edge}}(\phi),$$

with the canonical settings

$$\lambda_{\mathrm{gdd}} = 10^{-4}, \qquad \lambda_{\mathrm{edge}} = 1.$$

The scalar passed to L-BFGS is

$$F(\phi) = 10 \log_{10} J_{\mathrm{surf}}(\phi).$$

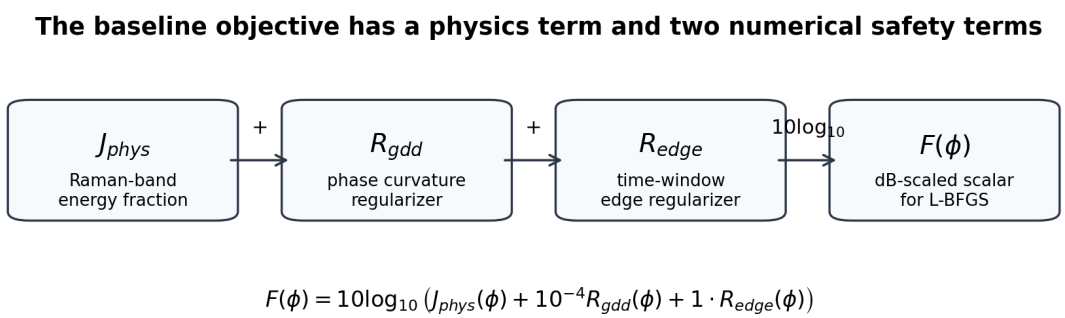


Figure 2: Baseline objective anatomy. The physics term measures Raman-band energy fraction. The two regularizers are numerical safety terms. The dB transform is applied after the full linear surface is assembled.

The gradient must match the same scalar surface:

$$\nabla_\phi F = \frac{10}{\ln 10} \frac{1}{J_{\mathrm{surf}}(\phi)} \nabla_\phi J_{\mathrm{surf}}.$$

This chain-rule scaling is important because the Raman fraction can become very small. The dB surface keeps the optimizer from losing useful gradient scale in deep-suppression regimes.

Intuition Check

The regularizers are not the main scientific result. They are guardrails. The main target is still "make the Raman-band fraction small." The guardrails discourage numerically suspicious phase curvature and pulses that touch the edge of the simulation time window.

4 Adjoint Gradient and L-BFGS

The high-dimensional part of the problem is the phase vector. With $N_t = 8192$, finite differences would be too expensive: one would need thousands of extra forward propagations to estimate a gradient.

The project instead uses an adjoint method. A forward solve gives the output field. A backward adjoint solve starts from the derivative of the output cost and propagates sensitivity back to the input. The phase-gradient form is

$$\frac{dF}{d\phi(\omega)} \propto \mathrm{Re}[\lambda_0(\omega)^* \, i \, u_\omega(0;\phi)] \, ,$$

with the proportionality including the dB scaling and regularizer gradients. L-BFGS then uses this gradient to update the phase without forming a full Hessian [6, 7].

Intuition Check

The adjoint is a shortcut for asking a very large "what if?" question. It answers, all at once, which phase pixels should move up or down to reduce the Raman-band output.

5 Implementation Surface

Component	Role in the baseline workflow
Canonical driver	Supported public entry point for the canonical single-mode optimization run.
Workflow wrapper	Dispatches approved run identifiers and output directories.
Optimization library	Defines the single-mode cost, adjoint-gradient wrapper, L-BFGS call, result payload, and trust report.
Problem setup	Builds the pulse, fiber, simulation grid, Raman-band mask, and shared simulation containers.
Standard-image helper	Saves the phase profile, phase diagnostic, optimized evolution heat map, and unshaped evolution heat map.
Cost-convention document	Defines the rule that regularizers are added before any optional dB transform.

The code names are intentionally kept out of the main story. For reproduction, the relevant paths are listed in the README next to this note.

6 Experimental Method

The maintained baseline experiment has five steps:

1. build the SMF-28 pulse, fiber, grid, and Raman-band mask;
2. evaluate the no-shaping control $\phi \equiv 0$;
3. run L-BFGS on the regularized dB objective;
4. save the standard image set for the shaped result and the unshaped control;
5. check determinism, boundary energy, photon conservation, gradient validation, and objective-surface metadata.

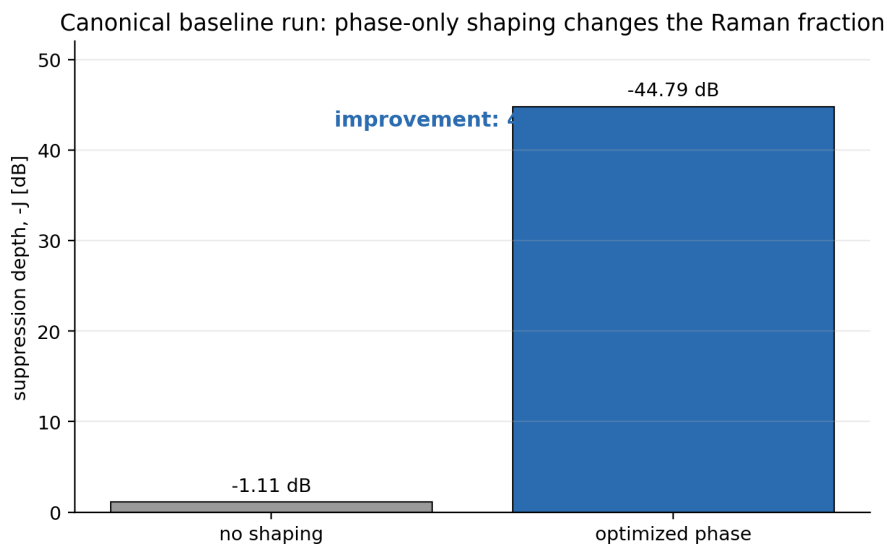


Figure 3: Canonical baseline scalar result. The no-shaping Raman-band fraction is $-1.11\,\mathrm{dB}$; the optimized phase reaches $-44.79\,\mathrm{dB}$, a $43.68\,\mathrm{dB}$ improvement.

Quantity	**Canonical run value**
Fiber and operating point	SMF-28, $L = 2\,\mathrm{m}$, $P = 0.2\,\mathrm{W}$
Grid	$N_t = 8192$, $27\,\mathrm{ps}$ time window
Input pulse	$185\,\mathrm{fs}$, centered at $1550\,\mathrm{nm}$
Initial objective	$-1.11\,\mathrm{dB}$
Final objective	$-44.79\,\mathrm{dB}$
Iterations	47
Converged flag	true

Numerical trust snapshot for the canonical baseline

determinism	FFTW/BLAS pinned	**PASS**
temporal boundary	max edge fraction 6.1e-6	**PASS**
photon conservation	drift 1.7e-6	**PASS**
gradient validation	max rel. error 2.4e-2	**PASS**
objective surface	regularized dB surface named	**PASS**
standard images	control plus shaped diagnostics	**PASS**

Figure 4: Numerical trust snapshot for the canonical run. The run passes the determinism, boundary, photon-conservation, gradient-validation, objective-surface, and standard-image gates.

Intuition Check

This is why the baseline is useful. It is not just a low number. It is a low number with a control group, a named objective, a gradient check, and pictures showing what phase was applied and what spectral evolution followed.

7 Representative Result Pages

The pages below pair each phase diagnostic with the corresponding spectral evolution heat map. This is the standard format for the rest of the note series.

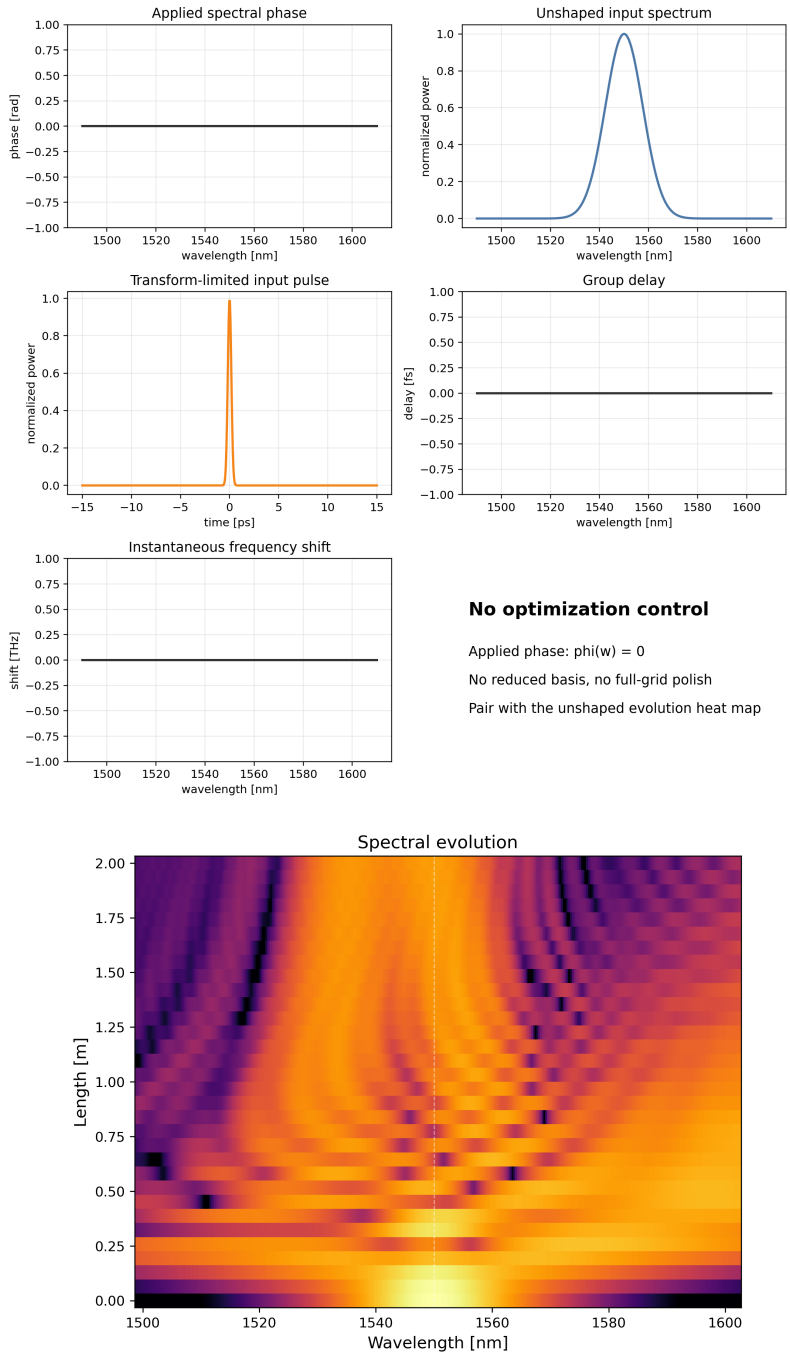
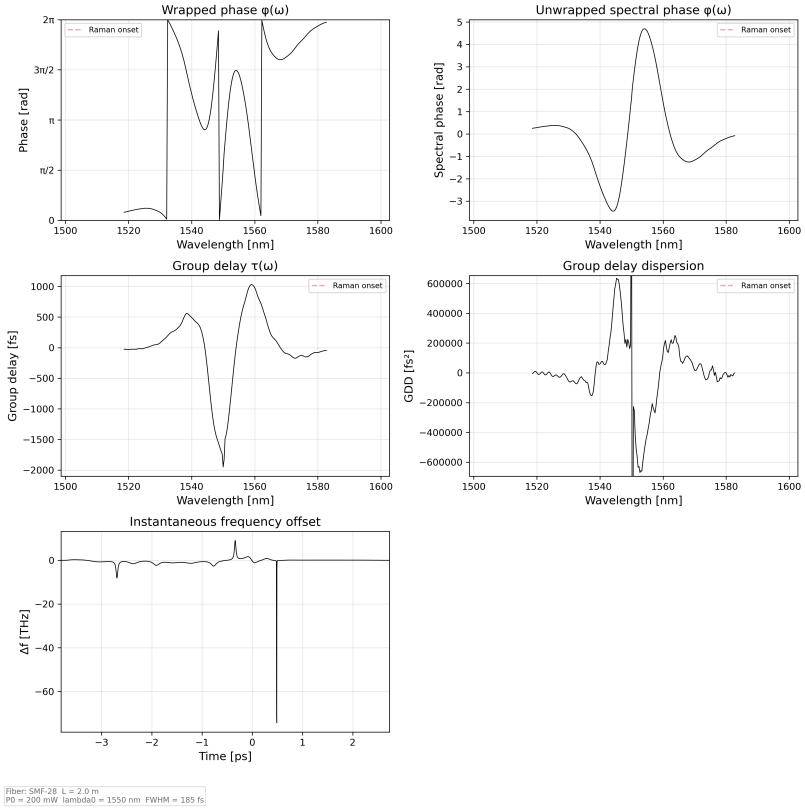


Figure 5: No-optimization control for the canonical SMF-28 baseline. The phase is flat, and the heat map shows the unshaped spectral evolution. This is the reference page used to judge whether the optimized phase changed the dynamics.



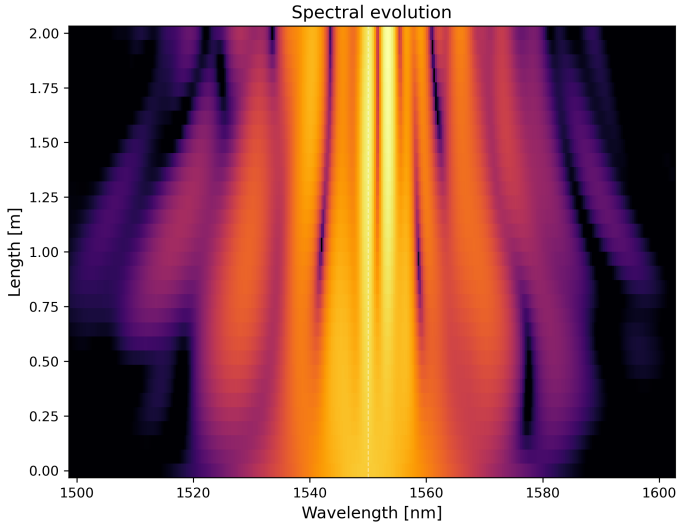
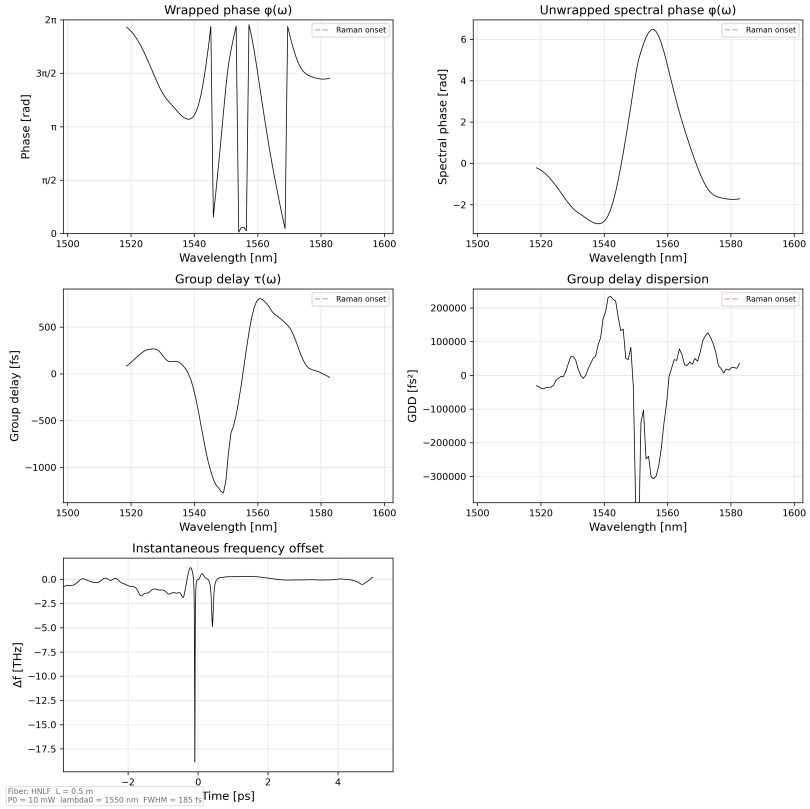


Figure 6: Optimized canonical SMF-28 baseline. The phase diagnostic shows the mask produced by L-BFGS; the heat map shows the spectral evolution under that mask. Read this page against the no-optimization control, not in isolation.



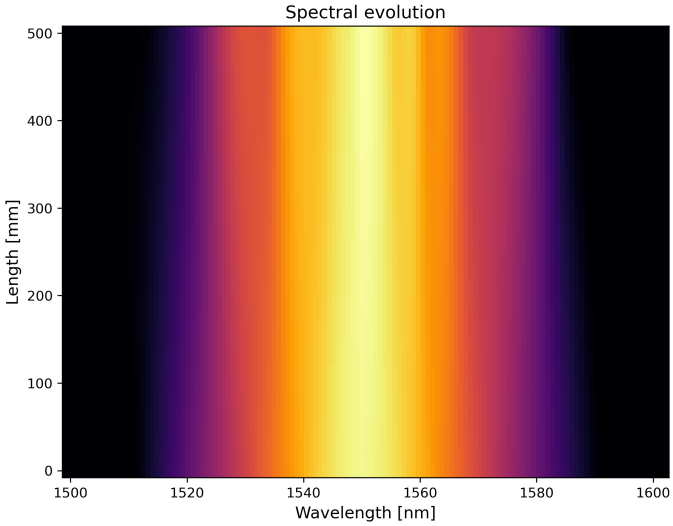


Figure 7: Reference single-mode HNLF run using the same phase-only diagnostic language. This is not the canonical SMF-28 baseline, but it shows that the same standard-image vocabulary transfers to another single-mode fiber setting.

8 How To Read The Images

Wrapped phase. This is $\phi(\omega)$ modulo 2π. It is close to what a phase modulator would implement.

Unwrapped phase. This removes 2π jumps so the broad phase shape is easier to see.

Group delay. This is the first derivative of phase with respect to frequency. It describes which wavelengths are delayed or advanced.

Group-delay dispersion. This is the derivative of group delay. Large spikes often indicate sharp phase structure and should be interpreted with care.

Spectral evolution heat map. The x-axis is wavelength and the y-axis is fiber length. Bright colors mean high spectral power. The important question is how the shaped heat map differs from the unshaped control.

Intuition Check

The phase diagnostic says what control was applied. The heat map says what the fiber did in response. A convincing result should show both.

9 Interpretation

The baseline establishes that phase-only shaping can strongly change the Raman-band output in the maintained single-mode setup. The canonical result is not meant to be a universal best mask. It is meant to be the clean reference case: one pulse, one fiber, one objective convention, one optimizer, one trust report, and one standard image set.

This also explains why later notes are necessary. A mask that is deeper may be less robust. A simpler mask may be easier to explain but shallower. A second-order optimizer may find a different basin. Those are not contradictions of the baseline; they are research directions built on top of it.

10 Presentation Capsule

If this note becomes the opening technical slide sequence, the clean message is:

- The project optimizes an input spectral phase so less output energy lands in the Raman-shifted band.
- The baseline result is a clear no-shaping versus optimized comparison: $-1.11\,\mathrm{dB}$ to $-44.79\,\mathrm{dB}$ at the maintained SMF-28 point.
- The standard image set is part of the evidence: phase profile, phase diagnostic, optimized heat map, and no-shaping heat map.
- Later notes should be explained as modifications of this baseline: different parameterizations, objectives, optimizers, fibers, or validation rules.

11 Limitations

Single-mode only. This note does not establish multimode behavior.

Phase-only only. It does not compare against amplitude or joint amplitude-phase shaping.

Not a global optimum. L-BFGS is local. The reported mask is a trustworthy baseline result, not proof that no better phase exists.

Objective-surface specific. The canonical run uses the regularized dB surface. Comparisons against physics-only linear costs, Hessian studies, or different regularizers must name the surface explicitly.

Images are necessary but not sufficient. A human-facing PDF should show the images, but the scalar metadata and trust checks still matter.

12 Reproduction Capsule

Canonical command:

```
julia -t auto -project=.  scripts/canonical/optimize_raman.jl
```

or, when available through the local workflow:

```
make optimize
```

Expected output is a timestamped directory under `results/raman/smf28_L2m_P0p2W_*` containing the result payload, JSON sidecar, numerical trust report, and the standard image set:

- phase profile image;
- phase diagnostic image;
- optimized spectral-evolution heat map;
- unshaped-control spectral-evolution heat map.

Substantial reruns should follow the project compute rules: use Julia with threading enabled, run heavy simulations on the burst machine when appropriate, and visually inspect the standard image set before calling the run complete.

13 Sources and References

References

[1] G. P. Agrawal. *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019. https://www.sciencedirect.com/book/9780128170427/nonlinear-fiber-optics

[2] J. M. Dudley, G. Genty, and S. Coen. "Supercontinuum generation in photonic crystal fiber." *Reviews of Modern Physics*, 78, 1135–1184, 2006. https://doi.org/10.1103/RevModPhys.78.1135

[3] J. Hult. "A fourth-order Runge-Kutta in the interaction picture method for simulating supercontinuum generation in optical fibers." *Journal of Lightwave Technology*, 25(12), 3770–3775, 2007. https://doi.org/10.1109/JLT.2007.909373

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[6] D. C. Liu and J. Nocedal. "On the limited memory BFGS method for large scale optimization." *Mathematical Programming*, 45, 503–528, 1989. https://doi.org/10.1007/BF01589116

[7] J. Nocedal and S. J. Wright. *Numerical Optimization*, 2nd ed. Springer, 2006. https://doi.org/10.1007/978-0-387-40065-5